SIM1003 Calculus II
Tutorial 4

All even-numbered questions in the following exercises

Exercise 9.1:

Convergence and Divergence

Which of the sequences an in Exercises 32 – 98 converge, and which diverge? Find the limit of each convergent sequence.

32. $a_n = \dfrac{n + (-1)^n}{n}$

33. $a_n = \dfrac{1 - 2n}{1 + 2n}$

34. $a_n = \dfrac{2n + 1}{1 - 3\sqrt{n}}$

35. $a_n = \dfrac{1 - 5n^4}{n^4 + 8n^3}$

36. $a_n = \dfrac{n + 3}{n^2 + 5n + 6}$

37. $a_n = \dfrac{n^2 - 2n + 1}{n - 1}, n \geq 2$

38. $a_n = \dfrac{1 - n^3}{70 - 4n^2}$

39. $a_n = 1 + (-1)^n$

40. $a_n = (-1)^n \left(1 - \dfrac{1}{n}\right)$

41. $a_n = \left(\dfrac{n + 1}{2n}\right)\left(1 - \dfrac{1}{n}\right)$

42. $a_n = \left(2 - \dfrac{1}{2^n}\right)\left(3 + \dfrac{1}{2^n}\right)$

43. $a_n = \dfrac{(-1)^{n+1}}{2n - 1}$

44. $a_n = \left(-\dfrac{1}{2}\right)^n$

45. $a_n = \sqrt{\dfrac{2n}{n + 1}}$

46. $a_n = \dfrac{1}{(0.9)^n}$

47. $a_n = \sin\left(\dfrac{\pi}{2} + \dfrac{1}{n}\right)$

48. $a_n = n\pi \cos(n\pi)$

49. $a_n = \dfrac{\sin n}{n}$

50. $a_n = \dfrac{\sin^2 n}{2^n}$

51. $a_n = \dfrac{n}{2^n}$

52. $a_n = \dfrac{3^n}{n^3}$

53. $a_n = \dfrac{\ln (n + 1)}{\sqrt{n}}$

54. $a_n = \dfrac{\ln n}{\ln 2n}$

55. $a_n = 8^{1/n}$

56. $a_n = (0.03)^{1/n}$

57. $a_n = \left(1 + \dfrac{7}{n}\right)^n$

58. $a_n = \left(1 - \dfrac{1}{n}\right)^n$

59. $a_n = \sqrt[n]{10n}$

60. $a_n = \sqrt[n]{n^2}$

61. $a_n = \left(\dfrac{3}{n}\right)^{1/n}$

62. $a_n = (n+4)^{1/(n+4)}$

63. $a_n = \dfrac{\ln n}{n^{1/n}}$

64. $a_n = \ln n - \ln(n+1)$

65. $a_n = \sqrt[n]{4^n n}$

66. $a_n = \sqrt[n]{3^{2n+1}}$

67. $a_n = \dfrac{n!}{n^n}$ (*Hint:* Compare with $1/n$.)

68. $a_n = \dfrac{(-4)^n}{n!}$

69. $a_n = \dfrac{n!}{10^{6n}}$

70. $a_n = \dfrac{n!}{2^n \cdot 3^n}$

71. $a_n = \left(\dfrac{1}{n}\right)^{1/(\ln\, n)}$

72. $a_n = \dfrac{(n+1)!}{(n+3)!}$

73. $a_n = \dfrac{(2n+2)!}{(2n-1)!}$

74. $a_n = \dfrac{3e^n + e^{-n}}{e^n + 3e^{-n}}$

75. $a_n = \dfrac{e^{-2n} - 2e^{-3n}}{e^{-2n} - e^{-n}}$

76. $a_n = \left(1 - \dfrac{1}{2}\right) + \left(\dfrac{1}{2} - \dfrac{1}{3}\right) + \left(\dfrac{1}{3} - \dfrac{1}{4}\right) + \cdots + \left(\frac{1}{n-2} - \frac{1}{n-1}\right) + \left(\frac{1}{n-1} - \frac{1}{n}\right)$

77. $a_n = (\ln 3 - \ln 2) + (\ln 4 - \ln 3) + (\ln 5 - \ln 4) + \cdots + (\ln(n-1) - \ln(n-2)) + (\ln n - \ln(n-1))$

78. $a_n = \ln\left(1 + \dfrac{1}{n}\right)^n$

79. $a_n = \left(\dfrac{3n+1}{3n-1}\right)^n$

80. $a_n = \left(\dfrac{n}{n+1}\right)^n$

81. $a_n = \left(\dfrac{x^n}{2n+1}\right)^{1/n}, x > 0$

82. $a_n = \left(1 - \dfrac{1}{n^2}\right)^n$

83. $a_n = \dfrac{3^n \cdot 6^n}{2^{-n} \cdot n!}$

84. $a_n = \dfrac{(10/11)^n}{(9/10)^n + (11/12)^n}$

85. $a_n = \tanh n$

86. $a_n = \sinh(\ln n)$

87. $a_n = \dfrac{n^2}{2n-1} \sin\dfrac{1}{n}$

88. $a_n = n\left(1 - \cos\dfrac{1}{n}\right)$

89. $a_n = \sqrt{n} \sin\dfrac{1}{\sqrt{n}}$

90. $a_n = (3^n + 5^n)^{1/n}$

91. $a_n = \tan^{-1} n$

92. $a_n = \dfrac{1}{\sqrt{n}} \arctan n$

93. $a_n = \left(\frac{1}{3}\right)^n + \frac{1}{\sqrt{2^n}}$

94. $a_n = \sqrt[n]{n^2 + n}$

95. $a_n = \frac{(\ln n)^{200}}{n}$

96. $a_n = \frac{(\ln n)^5}{\sqrt{n}}$

97. $a_n = n - \sqrt{n^2 - n}$

98. $a_n = \frac{1}{\sqrt{n^2 - 1} - \sqrt{n^2 + n}}$

Exercise 9.2:

Series with Geometric Terms

In Exercises 7–14, write out the first eight terms of each series to show how the series starts. Then find the sum of the series or show that it diverges.

7. $\sum_{n=0}^{\infty} \frac{(-1)^n}{4^n}$

8. $\sum_{n=2}^{\infty} \frac{1}{4^n}$

9. $\sum_{n=1}^{\infty} \left(1 - \frac{7}{4^n}\right)$

10. $\sum_{n=0}^{\infty} (-1)^n \frac{5}{4^n}$

11. $\sum_{n=0}^{\infty} \left(\frac{5}{2^n} + \frac{1}{3^n}\right)$

12. $\sum_{n=0}^{\infty} \left(\frac{5}{2^n} - \frac{1}{3^n}\right)$

13. $\sum_{n=0}^{\infty} \left(\frac{1}{2^n} + \frac{(-1)^n}{5^n}\right)$

14. $\sum_{n=0}^{\infty} \left(\frac{2^{n+1}}{5^n}\right)$

Using the *n*th-Term Test

In Exercises 31–38, use the *n*th-Term Test for divergence to show that the series is divergent, or state that the test is inconclusive.

31. $\sum_{n=1}^{\infty} \frac{n}{n + 10}$

32. $\sum_{n=1}^{\infty} \frac{n(n + 1)}{(n + 2)(n + 3)}$

33. $\sum_{n=0}^{\infty} \frac{1}{n + 4}$

34. $\sum_{n=1}^{\infty} \frac{n}{n^2 + 3}$

35. $\sum_{n=1}^{\infty} \cos \frac{1}{n}$

36. $\sum_{n=0}^{\infty} \frac{e^n}{e^n + n}$

37. $\sum_{n=1}^{\infty} \ln \frac{1}{n}$

38. $\sum_{n=0}^{\infty} \cos n\pi$

Telescoping Series

In Exercises 39-52, find a formula for the nth partial sum of the series and use it to determine whether the series converges or diverges. If a series converges, find its sum.

39. $\sum_{n=1}^{\infty}\left(\frac{1}{n}-\frac{1}{n+1}\right)$

40. $\sum_{n=1}^{\infty}\left(\frac{3}{n^2}-\frac{3}{(n+1)^2}\right)$

41. $\sum_{n=1}^{\infty}\left(\ln\sqrt{n+1}-\ln\sqrt{n}\right)$

42. $\sum_{n=1}^{\infty}(\tan\ (n)-\tan(n-1))$

43. $\sum_{n=1}^{\infty}\left(\arccos\left(\frac{1}{n+1}\right)-\arccos\left(\frac{1}{n+2}\right)\right)$

44. $\sum_{n=1}^{\infty}\left(\sqrt{n+4}-\sqrt{n+3}\right)$

45. $\sum_{n=1}^{\infty}\frac{4}{(4n-3)(4n+1)}$

46. $\sum_{n=1}^{\infty}\frac{6}{(2n-1)(2n+1)}$

47. $\sum_{n=1}^{\infty}\frac{40n}{(2n-1)^2(2n+1)^2}$

48. $\sum_{n=1}^{\infty}\frac{2n+1}{n^2(n+1)^2}$

49. $\sum_{n=1}^{\infty}\left(\frac{1}{\sqrt{n}}-\frac{1}{\sqrt{n+1}}\right)$

50. $\sum_{n=1}^{\infty}\left(\frac{1}{2^{1/n}}-\frac{1}{2^{1/(n+1)}}\right)$

51. $\sum_{n=1}^{\infty}\left(\frac{1}{\ln\,(n+2)}-\frac{1}{\ln\,(n+1)}\right)$

52. $\sum_{n=1}^{\infty}\left(\tan^{-1}(n)-\tan^{-1}(n+1)\right)$

Convergence or Divergence

Which series in Exercises 53–76 converge, and which diverge? Give reasons for your answers. If a series converges, find its sum.

53. $\sum_{n=0}^{\infty}\left(\frac{1}{\sqrt{2}}\right)^n$

54. $\sum_{n=0}^{\infty}\left(\sqrt{2}\right)^n$

55. $\sum_{n=1}^{\infty}(-1)^{n+1}\frac{3}{2^n}$

56. $\sum_{n=1}^{\infty}(-1)^{n+1}n$

57. $\sum_{n=0}^{\infty}\cos\left(\frac{n\pi}{2}\right)$

58. $\sum_{n=0}^{\infty}\frac{\cos n\pi}{5^n}$

59. $\sum_{n=0}^{\infty}e^{-2n}$

60. $\sum_{n=1}^{\infty}\ln\frac{1}{3^n}$

61. $\sum_{n=1}^{\infty}\frac{2}{10^n}$

62. $\sum_{n=0}^{\infty}\frac{1}{x^n},\ |x|>1$

63. $\sum_{n=0}^{\infty}\frac{2^n-1}{3^n}$

64. $\sum_{n=1}^{\infty}\left(1-\frac{1}{n}\right)^n$

65. $\sum_{n=0}^{\infty}\frac{n!}{1000^n}$

66. $\sum_{n=1}^{\infty}\frac{n^n}{n!}$

67. $\sum_{n=1}^{\infty}\frac{2^n+3^n}{4^n}$

68. $\sum_{n=1}^{\infty} \frac{2^n + 4^n}{3^n + 4^n}$

69. $\sum_{n=1}^{\infty} \ln\left(\frac{n}{n+1}\right)$

70. $\sum_{n=1}^{\infty} \ln\left(\frac{n}{2n+1}\right)$

71. $\sum_{n=0}^{\infty} \left(\frac{e}{\pi}\right)^n$

72. $\sum_{n=0}^{\infty} \frac{e^{n\pi}}{\pi^{ne}}$

73. $\sum_{n=1}^{\infty} \left(\frac{n}{n+1} - \frac{n+2}{n+3}\right)$

74. $\sum_{n=2}^{\infty} \left(\sin\left(\frac{\pi}{n}\right) - \sin\left(\frac{\pi}{n-1}\right)\right)$

75. $\sum_{n=1}^{\infty} \left(\cos\left(\frac{\pi}{n}\right) + \sin\left(\frac{\pi}{n}\right)\right)$

76. $\sum_{n=0}^{\infty} (\ln(4e^n - 1) - \ln(2e^n + 1))$

Exercise 9.3:

Determining Convergence or Divergence

Which of the series in Exercises 13–46 converge, and which diverge? Give reasons for your answers. (When you check an answer, remember that there may be more than one way to determine the series' convergence or divergence.)

13. $\sum_{n=1}^{\infty} \frac{1}{10^n}$

14. $\sum_{n=1}^{\infty} e^{-n}$

15. $\sum_{n=1}^{\infty} \frac{n}{n+1}$

16. $\sum_{n=1}^{\infty} \frac{5}{n+1}$

17. $\sum_{n=1}^{\infty} \frac{3}{\sqrt{n}}$

18. $\sum_{n=1}^{\infty} \frac{-2}{n\sqrt{n}}$

19. $\sum_{n=1}^{\infty} -\frac{1}{8^n}$

20. $\sum_{n=1}^{\infty} \frac{-8}{n}$

21. $\sum_{n=2}^{\infty} \frac{\ln n}{n}$

22. $\sum_{n=2}^{\infty} \frac{\ln n}{\sqrt{n}}$

23. $\sum_{n=1}^{\infty} \frac{2^n}{3^n}$

24. $\sum_{n=1}^{\infty} \frac{5^n}{4^n + 3}$

25. $\sum_{n=0}^{\infty} \frac{-2}{n+1}$

26. $\sum_{n=1}^{\infty} \frac{1}{2n-1}$

27. $\sum_{n=1}^{\infty} \frac{2^n}{n+1}$

28. $\sum_{n=1}^{\infty} \left(1 + \frac{1}{n}\right)^n$

29. $\sum_{n=2}^{\infty} \frac{\sqrt{n}}{\ln n}$

30. $\sum_{n=1}^{\infty} \frac{1}{\sqrt{n}(\sqrt{n}+1)}$

31. $\sum_{n=1}^{\infty} \frac{1}{(\ln 2)^n}$

32. $\sum_{n=1}^{\infty} \frac{1}{(\ln 3)^n}$

33. $\sum_{n=3}^{\infty} \frac{(1/n)}{(\ln n)\sqrt{\ln^2 n - 1}}$

34. $\sum_{n=1}^{\infty} \frac{1}{n\left(1 + \ln^2 n\right)}$

35. $\sum_{n=1}^{\infty} n \sin \frac{1}{n}$

36. $\sum_{n=1}^{\infty} n \tan \frac{1}{n}$

37. $\sum_{n=1}^{\infty} \frac{e^n}{1 + e^{2n}}$

38. $\sum_{n=1}^{\infty} \frac{2}{1 + e^n}$

39. $\sum_{n=1}^{\infty} \frac{e^n}{10 + e^n}$

40. $\sum_{n=1}^{\infty} \frac{e^n}{(10 + e^n)^2}$

41. $\sum_{n=2}^{\infty} \frac{\sqrt{n+2} - \sqrt{n+1}}{\sqrt{n+1}\sqrt{n+2}}$

42. $\sum_{n=3}^{\infty} \frac{7}{\sqrt{n+1}\,\ln\sqrt{n+1}}$

43. $\sum_{n=1}^{\infty} \frac{8 \tan^{-1} n}{1 + n^2}$

44. $\sum_{n=1}^{\infty} \frac{n}{n^2 + 1}$

45. $\sum_{n=1}^{\infty} \operatorname{sech} n$

46. $\sum_{n=1}^{\infty} \operatorname{sech}^2 n$